4.2.1 The four fundamental spaces of a matrix

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Week 04 due Feb 4, 2025 05:33 IST

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by a vector row vector, then what you're doing is you're taking linear

combinations of the rows of A. That of course is a row vector. The row space

of a consists of column vectors that can be created by taking linear combinations

of the rows of A when you view those rows as column vectors. That's what this

here says and obviously the row space of A is equal to the column space of the

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Reading assignment

1/1 point (graded)

Read Unit 4.2.1 of the notes. [LINK]

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Homework 4.2.1.1

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$. Show that its row space, $\mathcal{R}(A)$, and null space, $\text{Null}(A)$, are orthogonal.

- [x] done

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Homework 4.2.1.2

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$. Show that its column space, $\mathcal{C}(A)$, and left null space, $\text{Null}(A^H)$, are orthogonal.

- [x] done

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Homework 4.2.1.3

1/1 point (graded)
Let $\{s_0, \cdots, s_{r-1}\}$ be a basis for subspace $\mathcal{S} \subset \mathbb{C}^n$ and $\{t_0, \cdots, t_{k-1}\}$ be a basis for subspace $\mathcal{T} \subset \mathbb{C}^n$. Show that the following are equivalent statements:

- Subspaces $\mathcal{S}, \mathcal{T}$ are orthogonal.
- The vectors in $\{s_0, \cdots, s_{r-1}\}$ are orthogonal to the vectors in $\{t_0, \cdots, t_{k-1}\}$.
- $s_i^H t_j = 0$ for all $0 \leq i < r$ and $0 \leq j < k$.
- $\left(\begin{array}{c|c|c} s_0 & \cdots & s_{r-1} \end{array} \right)^H \left(\begin{array}{c|c|c} t_0 & \cdots & t_{k-1} \end{array} \right) = 0$, the zero matrix of appropriate size.

- [x] done

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Homework 4.2.1.4

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$. Show that any vector $x \in \mathbb{C}^n$ can be written as

Let $A \in \mathbb{C}$. Show that any vector $x \in \mathbb{C}$ can be written as $x = x_r + x_n$, where $x_r \in \mathcal{R}(A)$ and $x_n \in \mathrm{Null}(A)$, and $x_r^H x_n = 0$.

☑ done

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Solution:

For a complete solution, see the notes .

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Answers are displayed within the problem

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that the vector x can be uniquely written as the sum of a vector in the

row space, x_r, and a vector in the null space, x_n. And we know that the

vector x_n maps to the zero vector in C^m. Alright, so if we look at A times

x then we see that that's A times x_r plus x_n. And that means that it's A times

x_r, plus A times x_n. And we know that A times x_n is 0 . So we get that that's

equal to A times x_r. So what that really

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